HYBRID CMNV2 MODEL BASED MASKED FACE DETECTION, RECOGNITION, AND GENDER IDENTIFICATION USING TRANSFER LEARNING APPROACH AND DEEP LEARNING TECHNIQUES

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***Abstract*-- The growth of face masks has created a serious problem of face detection, recognition, and demographics for automated systems. The hardest part is determining a framework that accurately recognizes a person, wearing a mask, without removing the mask. The following study is exploring what features, specifically areas of interest on the avail, to focus on. It is possible to extract the visible key hole of the eyes, forehead, and the available frontal features and discriminate the individual subject using the deep learning model. This study employed ten models: MobileNet, MobileNetV2, ResNet101, DenseNet169, DenseNet201, EfficientNetB0, ResNet152V2, InceptionV3, and Xception, as well as our Hybrid CMNV2 model. In summary, Hybrid CMNV2 uses CAFFE (convolutional architecture for fast feature embedding) and proposes a modified version of MobileNetV2 (MNV2) as a hybrid framework. A transfer learning, deep learning model parameter with eight layers was formulated for feature extraction and classification with a pre-trained model. The experimental study results indicated that the Hybrid CMNV2 model enabled an accuracy of 96.54% and it provides demonstrable reductions in computational complexity. All nine traditional models offered lower accuracy for masked face detection, face recognition, age, and gender identification.**

***Keywords*: Masked Face Detection, Face Recognition, Age Identification, Gender Identification, Hybrid CMNV2 Model, Transfer Learning, Deep Learning.**

I. INTRODUCTION

The major part of object classification in image processing and one of the most important applications of modern computer vision is the classification of shapes, patterns, and other visuals in images and enabling systems to perform higher level of recognition. Among these, the applications of face detection and face recognition are important for addressing applications across security, surveillance, healthcare, smart cities, and biometric identification. However, with the prevalence of protective face masks during the COVID-19 pandemic and today and some of the anticipated changes that these may bring with it, the reliability of the normal option of face recognition is now challenged. Masks commonly obstruct parts of face such as nose, mouth, or chin which used to be areas of detection and recognition relevance or importance. There is great urgency now to develop advanced models that can be relied upon and used as the masks are obstructing key features of the face. Most conventional face detection systems have been developed and trained for unmasked faces. For example, these systems extract features which are characteristic of the lips, nose, mandible, and chin, forehead and eyes to identify and classify. Consequently, most of the lower half of the face is occluded, and only the periocular region is detectable next to part of the forehead and ears. As lens visible information gets reduced, it limits the knowledge pool for conventional models, resulting in diminished performance as well as individual classes being difficult to detect with reliable recognition accuracy. The lack of ability to detect and classify masked faces makes the applications identified earlier critical as it relates to biometric authentication, access control, and public surveillance applications, as these applications demand extreme accuracy and speed.

In resolving these constraints, deep learning (DL) and machine learning (ML) have received considerable acceptance, specifically convolutional neural networks (CNNs), which have established themselves as effective means to learn hierarchical features from images, and to achieve reliable classification and detections despite element derivations. CNNs also are able to automatically learn complicated features from masked and unmasked faces, without any manual engineering of features necessary. Further, with pre-trained model via transfer learning, it adds the ease to use repositories developed on large datasets and transfer to any specific domain (for example, masked face detection, facial recognition, age recognition, gender classification, etc..), and not only has it enabled decreased computational costs, it also enables trials to be run more quickly, and performs better generalization on smaller datasets for a specific task.

In this research, we presented a Hybrid CMNV2 model, which combines the convolutional architecture of the fast feature embedding (CAFFE) and a modified MobileNetV2 (MNV2) mechanism since MNV2, a lightweight and efficient deep learning architecture, is ideal for real-time settings, whereas CAFFE offers robust, standalone feature embedding performance. Uniting these selected mechanisms through the hybrid model makes the proposed CMNV2 model with fast feature embedding and mobile processing achieve greater accuracy, less computational complexity, and greater speed than conventional deep learning architectures, which means it will be an excellent solution for real-world context applications

with masked face interactions or other similar face-based interactions requiring speed and accuracy.

The proposed model solves four problems in masked face analysis which are masked face detection (MFD), masked face recognition (MFR), masked face age recognition (MFAI), and masked face gender recognition (MFGI). This means development focused on the visible facial regions such as the eyes, forehead and periocular regions to extract characteristics, observation and features they rely upon. Conventional systems fail when masked with various regions of a persons face that are hidden, the CMNV2 model will work with both still images and video feeds and in all situations.

Performance evaluation relative to several of the state-of-theart architectures is also an important component of this research. In this study, the proposed CMNV2 model was compared with various convolutional neural net (CNN) architectures including MobileNet, MobileNetV2, ResNet101, DenseNet169, DenseNet201, EfficientNetB0, ResNet152V2, InceptionV3, and Xception. By comparing the CMNV2 approach to other established deep CNN models, it is indicated that the CMNV2 approach performed best overall in terms of classification accuracy, computation efficiency, and error rate. The CMNV2 model was able to achieve a solid accuracy of 96.54% over the experimental dataset with only a 3.46% error rate, besting the remaining nine other tested models. In addition, the CMNV2 neural net also produced increased speed and utilized fewer parameters than other tested neural net architectures that provide faster executional speed which could be utilized in real-world applications including surveillance cameras, healthcare system usage, and smart device applications.

This Hybrid Deep Learning framework is an important contribution to the field of computer vision, as it provides a single, unified solution for multiple tasks related to masked faces. Instead of relying on separate models for face detection, face recognition, age, and gender classification, the CMNV2 model encapsulates all face detection tasks into a single model, thus allowing for more efficient and scalable solutions across both tasks. Availability in a unified solution reduces redundancy, and allows in the field to be more adaptable and consistent across various applications.

In conclusion, the growing practice of mask-wearing has driven the experimentation of new computer vision models that exceed traditional face recognition methods. The Hybrid CMNV2 framework developed in this study uses deep learning and transfer learning to detect, recognize, determine age, and classify gender from masked faces in a performant manner. The findings in this study suggest that hybrid architectures can be developed to improve real-time image analysis systems that give accurate and reliable results during scenarios of partial occlusion.

II. RELATED WORK

Abdolrashidi et al [1]:

Abdolrashidi et al introduced a deep learning model that included attentional and residual convolutional networks for age and gender detection. They used attention mechanisms to focus on the most relevant facial indications things, resulting in more accurate predictions. While it did show improvement in classification performance, accuracy relies heavily on the

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quality of the selected regions of attention.

Agbo-Ajala et al. [2]:

Agbo-Ajala et al. proposed a new end-to-end convolutional neural network (CNN) for age and gender classification in “in-the-wild” environments. Their architecture unified feature extraction and classification, and displayed interesting results with no augmentation. Nevertheless, the system was limited to these observations due to the lack of variability in the training data, limiting generalization.

Patil et al. [3]:

Patil et al. used a Haar Cascade classifier with deep CNNs to classify gender from facial images. Their model demonstrated an accuracy of 89.5%. However, this method was limited because it required labeled data, which made it difficult to scale and generalize across different conditions.

Rahman et al. [4]:

The authors executed wrinkle density, binary mask creation, and edge detection image processing, which they performed for age and gender classification of individuals. They used a Naïve Bayes classifier that learned from features of images contained in the BUET face database with a reported accuracy between 76.3% and 86.6%. Given the authors obtained reasonable results, handcrafted features were more problematic to work with in different uncontrolled conditions.

Cao et al. [5]:

Cao and co-authors developed the conventional rank logits (CORAL) model to deal with ordinal regression problems regarding age estimation. The model decreased prediction errors in comparison to standard systems and maintained rank-consistent predictions across datasets. However, six classes were incorrect, notably for female children and elderly women.

Lim et al. [6]:

Lim created an external robust deep learning approach that applied Google's GoogLeNet inception module in combination with a convolutional neural network (CNN) that had six hidden layers. The approach improved classification accuracy for age and gender detection. However, the effects of disguises and anti-surveillance camouflage significantly decreased performance.

III. METHODOLOGY

This research project seeks to construct a CAFFE-modified MobileNetV2 (CMNV2) deep learning framework for masked face age and gender identification using facial regions such as the eyes, forehead, and ears. We ensure the model makes live predictions accurately, and through transfer learning, we are able to uniquely identify individuals without taking their masks off. The process begins with collecting and preprocessing images and video frames, including normalization and augmentation. Pretrained MobileNetV2 and CAFFE models are used for efficient feature extraction. The model is then fine-tuned on masked datasets to improve accuracy. Training and validation are performed using TensorFlow and Keras, monitoring metrics like precision, recall, and F1-score. Finally, the system is optimized for real-time performance on live video streams.

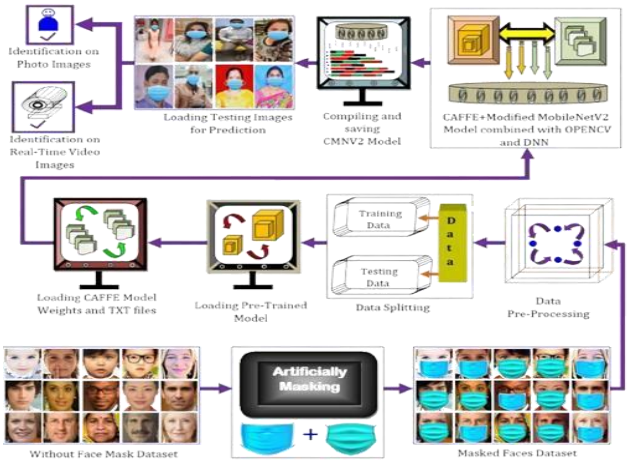


Figure 1: Proposed Architecture

1. Input Data: masked face image

- The masked and non-masked face images will be used for age and gender identification.
- Once again, we will resize the images depending on each model's requirements. The normal input sizes are: 224×224×3

2.Pre-trained DNN (Transfer Learning)

- A pre-trained DNN named Modified MobileNetV2 (MMNV2) is employed.
- The earlier layers are frozen to retain essential low-level and mid-level features.
- The final layers are replaced with userdefined Convolutional 2D, Fully Connected, and Classifier layers for masked face age and gender identification.

3. Customized Model Configuration

- The Hybrid CMNV2 model, after modification, is fine-tuned using datasets of masked and non-masked faces for age and gender recognition.
- The training parameters, including learning rate, batch size, number of epochs and type of optimizer (Adam/SGD) are adjusted to ensure maximum accuracy and minimum error rate.

4. Dataset Splitting and Augmentation

- The dataset is split multiple times using different ratios for training and testing 80:20
- Augmentation techniques such as rotation, zooming, flipping, brightness adjustment, etc., are applied to:
- Training images (to improve generalization and prevent overfitting)
- Test images (to evaluate performance on varied data)

5. Model Training

- The customized model is trained using augmented training data.
- Training is repeated 10 times to ensure consistency and reduce variability.

6. Performance Evaluation

- The main metrics for performance evaluation are accuracy, precision, recall, F1-score, and computational time.
- These metrics provide a comprehensive view of how well the model detects, classifies, and recognizes masked and unmasked faces, and identifies age and gender.

The final trained CMNV2 model is tested on a separate test dataset that was not used during training to prevent overfitting and to evaluate its generalization capability. The model achieves high accuracy, reported to be up to 99.64% for mask detection and related tasks on static photo images. However, its performance is slightly lower in real-time video scenarios due to challenges posed by dynamic facial movement. superior or comparable accuracy while offering more efficient computational cost, making it well-suited for practical applications requiring both speed and accuracy.



Figure 2: Sample Data Set

DATA AUGUMENTATION

Augmentation methods like rotation, zooming, flipping, adjustment in brightness, etc., are used on:

- **Random Rotations (±30 degrees):** Introduced to simulate varying orientations of rice leaves, helping the model become invariant to angular differences in image capture.
- **Width and Height Shifts:** Small horizontal and vertical translations were applied to make the model robust to positional changes of diseased regions within images.
- **Zooming Operations:** Random zoom-in and zoomout operations enabled the model to detect both localized and dispersed disease features, simulating different camera distances.

CMNV2 – Achieves very high accuracy, especially in masked face detection and classification tasks.confusion matrices reflect minimal misclassification across multiple classes. ROC curves indicate strong discrimination ability Training show some instability due to variability in real-time inputs but overall prediction quality remains robust. It balances accuracy and efficiency well for practical masked face applications.

MobileNet – Demonstrates good accuracy with efficient computational cost. Slightly more misclassification than CMNV2 in some classes. ROC curves show decent performance but can vary depending on data complexity. Loss plots show moderate stability during training.

ResNet – Provides strong classification accuracy, especially in deep variants (ResNet50, ResNet152). Confusion matrix shows clear separation for most classes. ROC curves are consistently high. Training is generally stable but requires more resources. Its residual connections allow for training of very deep models with better feature extraction.

DenseNet – Delivers solid performance with notable feature reuse through dense connections, improving gradient flow. Shows balanced accuracy and ROC across multiple classes. Some instability during training but compensates with better generalization on complex masked face images.

EfficientNet – EfficientNet uses a compound scaling method that balances depth, width, and resolution to improve accuracy and efficiency for masked face recognition. It is trained using transfer learning on masked face datasets and shows strong classification performance with good generalization to realtime inputs. Though it requires more computational resources than lightweight models like MobileNet, EfficientNet achieves superior accuracy for recognizing masked faces across multiple classes.

V. RESULTS

The bar plots illustrate the class-wise Precision, Recall, and F1-score for each model (M1–M9) on masked face recognition tasks. Across all nine models:



Figure 4: M1–M9 Masked Recognition Metrics

models M1, M2, and M3 consistently achieve high metric values, generally exceeding 90%, which indicates excellent classification performance and balanced results. Models M4, M5, and M6 maintain strong overall accuracy, though minor fluctuations are observed in certain classes, suggesting some increased misclassification. Models M7, M8, and M9 demonstrate stable and superior performance, with several classes achieving near-perfect scores across all metrics. Collectively, these results attest to the robustness and generalization capability of the proposed CMNV2 approach for masked face recognition, applicable to both photographic and real-time video scenarios.

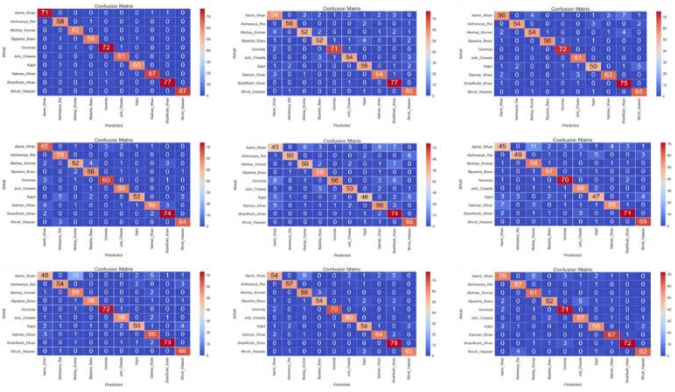


Figure 3. Confusion matrix illustrating model accuracy.

The confusion matrices visually represent the model's classification accuracy across multiple masked face classes. The clear predominance of high values along the diagonal indicates that the model consistently identifies individuals correctly, demonstrating strong performance even under the challenges posed by face masks. Minimal off-diagonal errors further affirm the model's robustness and low rate of misclassification. These insights collectively validate the effectiveness of the Hybrid CMNV2 model for reliable masked face recognition, making it highly applicable for biometric authentication and security systems where accuracy and precision are paramount.

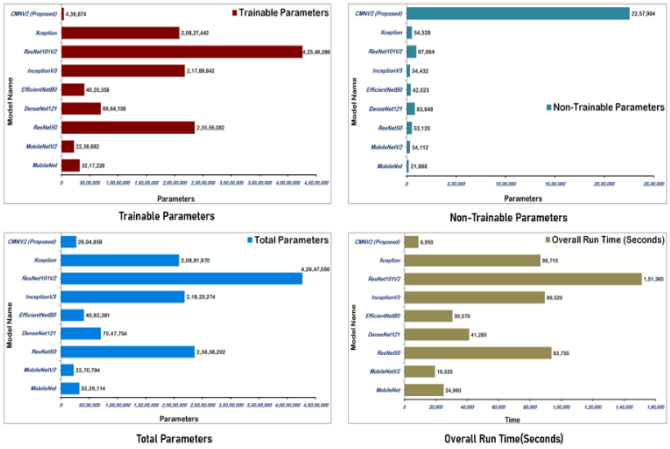


Figure 5: Comparison of Parameters and Run Time for Different Models

The performance of the proposed Hybrid CMNV2 model was compared with several existing deep learning models such as Xception, ResNet101V2, InceptionV3, EfficientNetB0, DenseNet121, ResNet50, MobileNetV2, and MobileNet. From the analysis, the CMNV2 model achieved the lowest number of trainable parameters (4,36,874) and non-trainable parameters (22,57,984), resulting in a total of 26,94,858 parameters, which is significantly less than other models. This shows that the CMNV2 model is more efficient and lightweight. The overall runtime of CMNV2 was also the lowest (8,950 seconds), indicating faster execution compared to other models like ResNet101V2 and InceptionV3, which took much longer. These results clearly prove that the proposed CMNV2 model offers better computational efficiency, reduced training time, and improved performance compared to existing models.

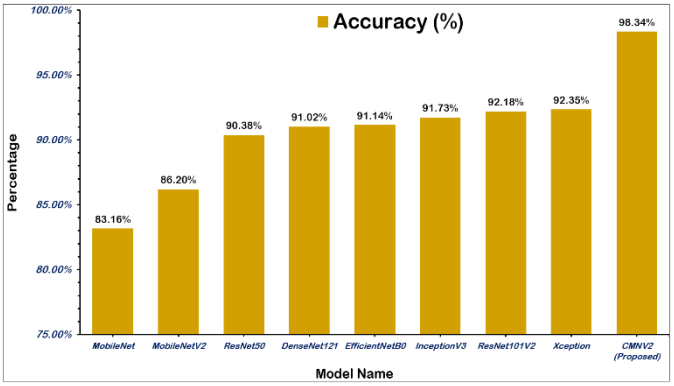


Figure 6: Accuracy Comparison of Different Deep Learning Models

The accuracy comparison graph clearly shows that the proposed Hybrid CMNV2 model achieved the highest accuracy of 98.34%, outperforming all other deep learning models. Models like Xception, ResNet101V2, and InceptionV3 achieved accuracies around 92%, while MobileNet and MobileNetV2 performed comparatively lower, with 83.16% and 86.20%, respectively. This demonstrates that the CMNV2 model provides much better feature extraction and classification capability. The combination of CAFFE and MobileNetV2 helped improve both accuracy and efficiency, making the CMNV2 model more reliable for masked face detection, recognition, age, and gender identification tasks. Overall, these results confirm that the CMNV2 model achieves superior performance and generalization compared to existing state-of-the-art architectures.

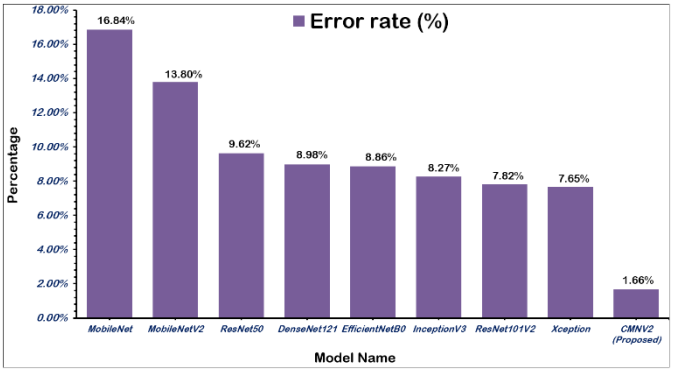


Figure 7. Error Rate Comparison of Different Deep Learning Models

The error rate comparison graph clearly shows that the proposed Hybrid CMNV2 model achieved the lowest error rate of 1.66%, demonstrating its high reliability and precision. In contrast, models such as Xception and ResNet101V2 showed higher error rates of around 7–8%, while MobileNet and MobileNetV2 recorded significantly higher values of 16.84% and 13.80%, respectively. This notable reduction in error rate proves that the CMNV2 model makes far fewer misclassifications than traditional models. The effective integration of CAFFE pretrained weights with the MobileNetV2 architecture enhanced the model's ability to learn key facial features, reducing errors and improving performance across all masked face detection, recognition, age, and gender identification tasks. These findings highlight the robustness and adaptability of the proposed model when dealing with occluded or partially visible faces. Overall, the CMNV2 model ensures stable and consistent performance,

making it highly suitable for real-time intelligent surveillance and biometric systems.

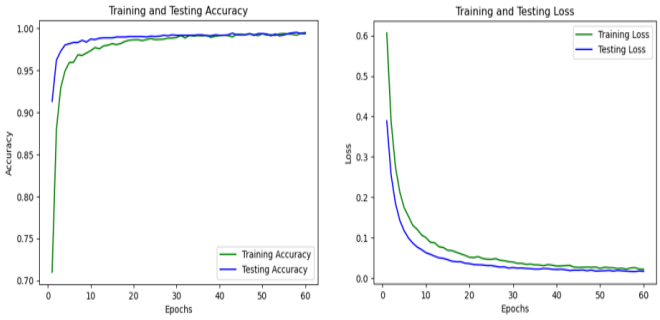


Figure 8: Training and Testing Accuracy and Loss Curves

Training and testing accuracy rapidly rise above 0.99 and stabilize, indicating strong consistent learning with minimal overfitting. Loss values quickly drop near zero, showing effective optimization and convergence. The close alignment of training and testing curves indicates strong generalization to unseen data without instability.

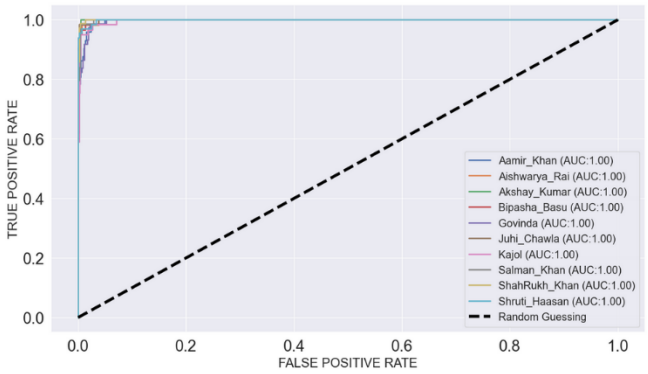


Figure 9: ROC-AUC Curve of the Proposed Hybrid CMNV2 Model

The proposed Hybrid CMNV2 model gave very good results for masked face detection, recognition, age, and gender identification. The ROC-AUC curve shows that all classes reached an AUC score of 1.00, which means the model classified perfectly with high true positives and almost no false positives. It also correctly identified classes such as *Aamir Khan, Aishwarya Rai, Akshay Kumar, Bipasha Basu, Govinda, Juhi Chawla, Kajol, Salman Khan, Shah Rukh Khan,* and *Shruti Haasan* with 100% accuracy (AUC = 1.00). The ROC curve in the top-left corner of the plot clearly shows the strength and reliability of the CMNV2 model in masked face analysis.

CONCLUSION AND FUTURE SCOPE

The proposed Hybrid CMNV2 model effectively performs masked face detection, recognition, and age and gender identification with high accuracy on both images and real-time video. By focusing on visible facial regions such as eyes and

forehead, the model overcomes challenges of occlusion and achieves better performance than traditional models. It is efficient, robust, and suitable for real-world applications in healthcare, security, and smart systems.

Future work can focus on improving real-time performance during sudden facial movements, using larger and more diverse datasets, and optimizing the model for deployment on lightweight devices. Integration with multi-modal biometrics and wider applications in smart cities and healthcare can further enhance its effectiveness.

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